Mattia Giurato, Ph.D.

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Areas of Specialisation

- **GNC & AOCS** — Attitude determination and control (ADCS/AOCS) for spacecraft; orbit GNC; FDIR logic design compliant with ECSS and Savoir guidelines; model-based design with MATLAB/Simulink and automatic code generation for flight software.
- **UAV Systems** — Multirotor and fixed-wing UAV design, identification and control; fully-actuated and over-actuated platform architectures; PX4/Ardupilot firmware customisation; localisation in GPS-denied environments (motion-capture, stereo-vision, optical flow).
- **Control Theory** — Robust control (structured $\mathcal{H}_\infty$ synthesis), nonlinear control, adaptive control, control allocation for over-actuated systems.
- **Estimation** — System identification; parameter estimation; attitude estimation (complementary filter, EKF); sensor fusion (IMU, GPS, optical flow).
- **Verification & Validation** — Full V&V pyramid: MIL, SIL, PIL, and HIL test campaigns; requirements management and flowdown.

Education

- 11/16 – 01/20 — **Ph.D. in Aerospace Engineering**, Politecnico di Milano, Italy.
  Advisor: Prof. Marco Lovera.
  Thesis: *Design, Integration, and Control of Multirotor UAV Platforms*.
- 10/13 – 12/15 — **M.Sc. in Automation and Control Engineering**, Politecnico di Milano, Italy.
  Main topics: Robust control, Nonlinear control, Adaptive control, Estimation.
  Thesis: *Design, Integration, and Control of a Multirotor UAV Platform*.
- 09/10 – 10/13 — **B.Sc. in Automation and Control Engineering**, Politecnico di Milano, Italy.

Teaching & Research Experience

- 02/20 – 10/20 — **Post-Doctoral Researcher**, Department of Aerospace Science and Technologies, Politecnico di Milano, Italy.
  Research topics: multirotor GNC, attitude and position control, model identification, parameter estimation, state estimation (EKF, complementary filter). (*Concurrent with ANT-X s.r.l. until full-time transition in 10/20.*)
- 03/19 – 07/19 — **Visiting Researcher**, LAAS – CNRS, Toulouse, France.
  Research topics: control allocation of fully-actuated and over-actuated multirotors.
- 2018 – 2019 — **Teaching Assistant**, Department of Aerospace Science and Technologies, Politecnico di Milano, Italy.
  Course: *Estimation in Aerospace* (M.Sc. tracks in Aeronautics and Space Engineering). Responsibilities: lecture support, exercise sessions, laboratory activities.
- **Course Organiser and Lecturer**, *UAV Lab* – Innovative Teaching Methods course, Drone Laboratory, Department of Aerospace Science and Technologies, Politecnico di Milano, Italy.
  Multidisciplinary course on the design and integration of a multirotor UAV; topics covered: aeromechanics, electronics, control, and flight testing.

Teaching & Research Experience (continued)

01/16 – 11/16 **Research Assistant**, Department of Aerospace Science and Technologies, Politecnico di Milano, Italy.
Research activities: quadrotor attitude identification and control, on-board attitude estimators, robust $\mathcal{H}_\infty$ control law design, development of MATLAB/Simulink simulation environments, UAV design and integration.

2014 **Course Organiser and Lecturer**, *Aerospace System, Guidance, Navigation, and Control* – course organised in collaboration between Skyward Experimental Rocketry and The MathWorks.

Research Publications

Journal Articles

1. M. Ryll, D. Bicego, **M. Giurato**, M. Lovera, and A. Franchi, "Fast-hex – a morphing hexarotor: Design, mechanical implementation, control and experimental validation," *IEEE/ASME Transactions on Mechatronics*, vol. 27, no. 3, pp. 1244–1255, 2022. DOI: 10.1109/TMECH.2021.3099197
2. D. Invernizzi, **M. Giurato**, P. Gattazzo, and M. Lovera, "Comparison of control methods for trajectory tracking in fully actuated unmanned aerial vehicles," *IEEE Transactions on Control Systems Technology*, vol. 29, no. 3, pp. 1147–1160, 2021. DOI: 10.1109/TCST.2020.2992389
3. G. Bressan, A. Russo, D. Invernizzi, **M. Giurato**, S. Panza, and M. Lovera, "Adaptive augmentation of the attitude control system for a multirotor unmanned aerial vehicle," *Proceedings of the Institution of Mechanical Engineers, Part G: Journal of Aerospace Engineering*, vol. 234, no. 10, pp. 1587–1596, 2020. DOI: 10.1177/0954410018820193
4. G. Gozzini, D. Invernizzi, S. Panza, **M. Giurato**, and M. Lovera, "Air-to-air automatic landing of unmanned aerial vehicles: A quasi time-optimal hybrid strategy," *Journal of Guidance Control and Dynamics*, vol. 4, no. 3, pp. 692–697, 2020. DOI: 10.1109/LCSYS.2020.2991701

Book Chapters

1. D. Del Cont Bernard, F. Riccardi, **M. Giurato**, and M. Lovera, "Ground effect analysis for a quadrotor platform," in *Advances in Aerospace Guidance, Navigation and Control*, Springer, 2018.

Conference Proceedings

1. **M. Giurato**, D. Invernizzi, S. Panza, and M. Lovera, "The role of laboratory activities in aerospace control education: Two case studies," in *Proceedings of the 21st IFAC World Congress*, Berlin, Germany, 2020.
2. P. Giuri, A. Marini Cossetti, **M. Giurato**, D. Invernizzi, and M. Lovera, "Air-to-air automatic landing for multirotor uavs," in *Proceedings of the 5th CEAS Conference on Guidance, Navigation and Control*, Milano, Italy, 2019.
3. **M. Giurato**, S. Panza, and M. Lovera, "Robust filtering for very high accuracy attitude determination," in *Proceedings of the 58th Israel Annual Conference on Aerospace Sciences*, Tel Aviv and Haifa, Israel, 2018.
4. D. Invernizzi, **M. Giurato**, P. Gattazzo, and M. Lovera, "Full pose tracking for a tilt-arm quadrotor uav," in *Proceedings of the IEEE Conference on Control Technology and Applications*, Copenhagen, Denmark, 2018.
5. D. Del Cont Bernard, **M. Giurato**, F. Riccardi, and M. Lovera, "Ground effect analysis for a quadrotor platform," in *Proceedings of the 4th CEAS Specialist Conference on Guidance, Navigation & Control*, Warsaw, Poland, 2017.
6. F. Haydar, **M. Giurato**, M. Lovera, and G. Sechi, "Very high accuracy attitude determination for los steering," in *Proceedings of the 10th International ESA Conference on Guidance, Navigation & Control Systems*, Salzburg, Austria, 2017.

7. A. Russo, D. Invernizzi, **M. Giurato**, and M. Lovera, "Adaptive augmentation of the attitude control system for a multirotor uav," in *Proceedings of the 7th European Conference for Aeronautics and Space Sciences*, Milano, Italy, 2017.
8. **M. Giurato** and M. Lovera, "Quadrotor attitude determination: A comparison study," in *Proceedings of the 2016 IEEE Conference on Control Applications (CCA)*, Buenos Aires, Argentina, 2016.
9. A. Rivolta, **M. Giurato**, F. Cuzzocrea, F. Rovere, and S. Farí, "Low cost mems imu calibration for aerospace student activities," in *Proceedings of the 1st Symposium On Space Educational Activities*, 2015.

Thesis Co-Supervision

2020

- N. Battaini. *Design and Dynamic Modeling of a VTOL UAV*. Aeronautics Engineering, M.Sc., Politecnico di Milano, 2020. [link]
- K. Bstaoros. *Kalman Filtering for UAV Navigation Using Optical Flow*. Aeronautics Engineering, M.Sc., Politecnico di Milano, 2020. [link]
- L. Carcione. *Attitude and Position Estimation for a 2 DoF Drone*. Aeronautics Engineering, M.Sc., Politecnico di Milano, 2020. [link]
- W. Nguemen Djonkoua. *Model-Reference Adaptive Control of the Vertical Dynamics of a Multirotor UAV*. Aeronautics Engineering, M.Sc., Politecnico di Milano, 2020. [link]

2019

- A. Zangarini. *Data-Driven Multivariable Attitude Control Design for Multirotor UAV Platforms*. Aeronautics Engineering, M.Sc., Politecnico di Milano, 2019. [link]
- U. Arshad. *Adaptive Control Implementation to Include Ground Effect on UAV Simulator*. Aeronautics Engineering, M.Sc., Politecnico di Milano, 2019. [link]
- G. Gozzini. *UAV Autonomous Landing on a Moving Aerial Vehicle*. Aeronautics Engineering, M.Sc., Politecnico di Milano, 2019. [link]
- A. Gatti. *Modelling, Identification and Control of a Fixed-Wing UAV*. Aeronautics Engineering, M.Sc., Politecnico di Milano, 2019. [link]
- D. Migliore. *Structured Model Identification for Multirotor UAVs*. Aeronautics Engineering, M.Sc., Politecnico di Milano, 2019. [link]
- G. Roggi. *Vision-Based Pilot Support System for Multirotor PV Plant Inspection*. Aeronautics Engineering, M.Sc., Politecnico di Milano, 2019. [link]

2018

- S. Musacchio. *Optimal and Robust UAV State Estimation Based on GPS and Optical Flow*. Aeronautics Engineering, M.Sc., Politecnico di Milano, 2018. [link]
- P. Giuri, A. Marini Cossetti. *Air-to-Air Automatic Landing for Multirotor UAVs*. Aeronautics Engineering, M.Sc., Politecnico di Milano, 2018. [link]
- M. Maccotta. *Multirotor UAVs for Fugitive Emissions Detection: Sizing, Modelling and Control System Design*. Automation and Control Engineering, M.Sc., Politecnico di Milano, 2018. [link]
- E. Balcioglu. *Development of a Conceptual Design Tool for Multirotors*. Aeronautics Engineering, M.Sc., Politecnico di Milano, 2018. [link]
- G. Bressan. *Hardware/Software Architecture, Code Generation and Control for Multirotor UAVs*. Computer Science and Engineering, M.Sc., Politecnico di Milano, 2018. [link]

2017

- P. Gattazzo. *Nonlinear Control of a Tilt-Arm Quadrotor UAV*. Automation and Control Engineering, M.Sc., Politecnico di Milano, 2017. [link]
- D. Carelli. *Nonlinear Attitude and Position Control for a Quadrotor UAV*. Automation and Control Engineering, M.Sc., Politecnico di Milano, 2017. [link]

Thesis Co-Supervision (continued)

- A. Delbono. *Consensus Based Control for an Unmanned Aerial Vehicle Formation*. Computer Science and Engineering, M.Sc., Politecnico di Milano, 2017. [link]
- D. Chevallard. *Design, Identification and Control of a Micro Aerial Vehicle*. Aeronautics Engineering, M.Sc., Politecnico di Milano, 2017. [link]
- R. Salbati. *Ground Effect Compensation for Multicopter UAV*. Exchange student from Université de Liège (ULg), Aeronautics Engineering, M.Sc., Politecnico di Milano, 2017.
- S. Farì. *Guidance and Control for a Fixed-Wing UAV*. Automation and Control Engineering, M.Sc., Politecnico di Milano, 2017. [link]

2016

- M. Ferronato. *Identificazione e Controllo della Dinamica Verticale di un Elicottero Quadrirotore*. Automation and Control Engineering, M.Sc., Politecnico di Milano, 2016. [link]
- C. Micheli. *Design, Identification and Control of a Tiltrotor Quadcopter UAV*. Automation and Control Engineering, M.Sc., Politecnico di Milano, 2016. [link]
- D. Di Bacco. *Distance Control from Vertical Surfaces of a Multirotor UAV Designed for Structural Health Monitoring of Civil Infrastructures*. Aeronautics Engineering, M.Sc., Politecnico di Milano, 2016. [link]
- D. Del Cont Bernard. *Ground Effect Analysis for a Quadrotor Platform*. Aeronautics Engineering, M.Sc., Politecnico di Milano, 2016. [link]
- A. De Angelis, A. Sorbelli. *Software Architecture, Estimators and Control for Multirotor UAVs*. Computer Science and Engineering, M.Sc., Politecnico di Milano, 2016. [link]
- A. Russo. *Adaptive Control of Multirotor UAVs*. Automation and Control Engineering, M.Sc., Politecnico di Milano, 2016. [link]
- T. Chupin. *Data-Driven Attitude Control Design for Multirotor UAVs*. Automation and Control Engineering, M.Sc., Politecnico di Milano, 2016. [link]